PROJECT IDEA: THE E-COMMERCE CUSTOMER EXPERIENCE INVESTIGATION

If you're building your first serious Data Analytics project,
don't build another "Sales Dashboard".

Build a project where you investigate a REAL business problem.

DATASET:
Olist Brazilian E-Commerce Public Dataset

Source:
Kaggle → Olist Brazilian E-Commerce Public Dataset

Dataset link:
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

BUSINESS SCENARIO

You are a Data Analyst at an e-commerce company.

The company is getting thousands of orders, but leadership
wants to understand:

"WHAT ACTUALLY DRIVES A CUSTOMER TO HAVE A GOOD OR BAD
PURCHASING EXPERIENCE?"

You have access to:

- Orders
- Customers
- Sellers
- Products
- Payments
- Delivery information
- Customer reviews
- Geographic data

Your job is to investigate the entire customer journey.

The goal is NOT just to report sales.

The goal is to discover:

WHAT DRIVES CUSTOMER SATISFACTION,
WHAT CAUSES BAD REVIEWS,
AND WHERE THE BUSINESS CAN IMPROVE.

YOUR MAIN BUSINESS QUESTIONS

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1. What factors are most strongly associated with low customer satisfaction?

2. Does late delivery lead to poor reviews?

3. Which product categories receive the worst reviews?

4. Which sellers consistently underperform?

5. Are expensive products more likely to receive negative reviews?

6. Does delivery distance affect customer satisfaction?

7. Which cities/regions have the worst delivery experience?

8. Which product categories have high revenue but poor ratings?

9. Which sellers generate high revenue but poor customer experience?

10. Are customers with negative experiences less likely to purchase again?

11. What are the biggest themes in negative customer reviews?

12. What should the business do to improve customer experience?

PHASE 1 — UNDERSTAND THE DATA

Explore the available tables.

Main tables include:

• orders
• order_items
• customers
• sellers
• products
• payments
• reviews
• geolocation

Understand:

→ Primary keys
→ Foreign keys
→ Relationships
→ Missing values
→ Duplicate records
→ Data types
→ Date ranges

PHASE 2 — DATA CLEANING

Use Python + Pandas.

Perform:

• Missing-value analysis
• Duplicate detection
• Date conversion
• Category standardization
• Outlier detection

• Data-type correction
• Referential-integrity checks

Create a clean analytical dataset by joining the relevant tables.

PHASE 3 — SQL ANALYSIS

Use SQL to investigate:

REVENUE:

• Total revenue
• Monthly revenue
• Average order value
• Revenue by product category
• Revenue by seller
• Revenue by state

CUSTOMER EXPERIENCE:

• Average review score
• Review score by category
• Review score by seller
• Review score by state
• Review score by order value

DELIVERY:

• Estimated vs actual delivery time
• Average delivery delay
• Late-delivery percentage
• Delivery performance by seller
• Delivery performance by geography

BUSINESS PERFORMANCE:

• High-revenue / low-rating sellers
• High-revenue / low-rating categories
• High-volume / poor-delivery regions
• Sellers with consistently poor performance

PHASE 4 — CREATE NEW METRICS

Don't rely only on the columns provided.

Create analytical metrics such as:

DELIVERY DELAY

Actual delivery date
- Estimated delivery date

LATE DELIVERY FLAG

Actual delivery date > Estimated delivery date

FREIGHT RATIO

Freight value / Product price

CUSTOMER EXPERIENCE SCORE

Based on:

• Review score
• Delivery performance
• Review sentiment

SELLER PERFORMANCE SCORE

Based on:

• Review score
• Delivery performance
• Order volume

PHASE 5 — PYTHON EXPLORATORY ANALYSIS

Use Python to investigate relationships such as:

Delivery delay
↓
Review score

Order value
↓
Review score

Freight ratio
↓
Review score

Product category
↓
Customer satisfaction

Seller performance
↓
Customer satisfaction

Don't just create charts.

For every chart ask:

"SO WHAT?"

PHASE 6 — THE INTERESTING PART: CUSTOMER REVIEWS + AI

You have thousands of customer review comments.

Instead of manually reading them, use an LLM.

For every review, extract:

• Sentiment
• Issue category
• Severity
• Main complaint
• Root cause
• Positive/negative theme

Example:

Review:

"The delivery was extremely late and the product arrived damaged."

AI classification:

Sentiment:
Negative

Issue:
Delivery + Product Damage

Severity:
High

Potential Root Cause:
Logistics/Handling

PHASE 7 — BUILD A CUSTOMER COMPLAINT TAXONOMY

Group negative reviews into themes:

• Late delivery
• Product damage
• Wrong product
• Missing product
• Product quality
• Seller issue
• Packaging
• Shipping
• Payment
• Other

Then answer:

"What are customers actually complaining about?"

PHASE 8 — AI HYPOTHESIS GENERATION

Give the initial findings to an AI tool.

Ask:

"Based on these findings, generate possible explanations
for poor customer satisfaction."

Possible hypotheses:

H1:
Late delivery is a major driver of negative reviews.

H2:
Certain sellers have systematically poor customer experience.

H3:
Some product categories have unusually high complaint rates.

H4:
High freight costs are associated with poor customer satisfaction.

H5:
Certain geographic regions have operational problems.

IMPORTANT:

AI DOES NOT GIVE YOU THE ANSWER.

AI GENERATES HYPOTHESES.

YOU TEST THEM WITH DATA.

PHASE 9 — AI VS DATA

Create a framework:

HYPOTHESIS
↓
DATA TEST
↓
STATISTICAL EVIDENCE
↓
VERDICT

Example:

Hypothesis:
Late deliveries cause lower review scores.

Test:

Compare review scores for on-time vs late orders.

Result:
Calculate the difference.

Statistical test:
Test whether the difference is meaningful.

Verdict:
Confirmed / Rejected / Inconclusive

This is a VERY important part of the project.

It demonstrates that you don't blindly trust AI.

PHASE 10 — STATISTICAL ANALYSIS

Choose 1–2 important hypotheses.

Example:

"Does late delivery significantly affect customer ratings?"

Define:

H0:
Late delivery has no significant relationship with
customer rating.

H1:
Late delivery is associated with lower customer ratings.

Use an appropriate statistical test.

Report:

• Test used

• Sample size
• p-value
• Effect size
• Confidence interval
• Business interpretation

PHASE 11 — SELLER ANALYSIS

Build a seller performance framework.

For every seller calculate:

• Orders
• Revenue
• Average rating
• Late-delivery rate
• Average delivery delay
• Freight ratio
• Negative-review rate

Then create segments:

HIGH REVENUE + HIGH EXPERIENCE
→ Best performers

HIGH REVENUE + LOW EXPERIENCE
→ PRIORITY PROBLEM

LOW REVENUE + HIGH EXPERIENCE
→ Growth opportunity

LOW REVENUE + LOW EXPERIENCE
→ Low priority

PHASE 12 — PRODUCT CATEGORY ANALYSIS

Analyze:

• Revenue
• Orders
• Average rating
• Negative-review rate
• Delivery performance
• Freight ratio

Find categories that are:

HIGH REVENUE + LOW SATISFACTION

These are especially important because improving them could have a significant business impact.

PHASE 13 — GEOGRAPHIC ANALYSIS

Analyze customer experience by:

• State
• City
• Region

Compare:

• Orders
• Revenue
• Delivery delay
• Review score
• Freight ratio

Find geographic hotspots where the customer experience is significantly worse.

PHASE 14 — CUSTOMER JOURNEY ANALYSIS

Map the journey:

CUSTOMER
↓
ORDER
↓
SELLER
↓
PRODUCT
↓
DELIVERY
↓
CUSTOMER REVIEW

Ask:

"At which stage does the experience break down?"

PHASE 15 — POWER BI DASHBOARD

Create 5 dashboard pages.

PAGE 1 — EXECUTIVE OVERVIEW

Show:

• Revenue
• Orders
• Average rating
• Late-delivery rate
• Negative-review rate

Main question:

"How healthy is the customer experience?"

PAGE 2 — DELIVERY EXPERIENCE

Show:

- On-time vs late deliveries
- Average delivery delay
- Delivery performance by seller
- Delivery performance by region
- Delivery vs review score

PAGE 3 — CUSTOMER VOICE

Show:

- Sentiment
- Complaint categories
- Top complaint themes
- Negative-review trends
- Severity

PAGE 4 — SELLER & PRODUCT PERFORMANCE

Show:

- Seller performance
- Category performance
- Revenue vs rating
- Revenue vs delivery performance

PAGE 5 — ROOT CAUSE & RECOMMENDATIONS

Show:

- Biggest customer pain points
- Root causes

• Priority areas
• Recommended actions
• Expected impact

PHASE 16 — FINAL BUSINESS INSIGHTS

Your project should end with 5–7 concrete findings.

Example format:

FINDING:

Late deliveries are strongly associated with lower customer ratings.

EVIDENCE:

X% of late orders received ratings below Y.

BUSINESS IMPACT:

Poor delivery performance is concentrated in specific regions/sellers.

RECOMMENDATION:

Prioritize operational improvements for those sellers/regions.

Do this for every major finding.

PHASE 17 — RECOMMENDATIONS

Don't write generic recommendations like:

"Improve customer service."

Instead:

"Prioritize sellers with high order volume and consistently poor delivery performance because they affect a larger number of customers."

Or:

"Investigate categories with high revenue but disproportionately high negative-review rates."

Every recommendation should contain:

ACTION
+
WHY
+
EXPECTED IMPACT
+
HOW TO MEASURE IT

FINAL DELIVERABLES

Your portfolio should contain:

1. SQL scripts

2. Python notebook

3. AI/NLP analysis

4. Statistical analysis

5. Power BI dashboard

6. Executive presentation

7. Business recommendations

8. GitHub README

TOOLS

SQL
→ PostgreSQL / MySQL

Python
→ Pandas
→ NumPy
→ Matplotlib
→ Seaborn
→ SciPy / Statsmodels

BI
→ Power BI

AI
→ ChatGPT / Claude / Gemini

Version Control
→ GitHub

WHY THIS IS A STRONG FRESHER PROJECT

You aren't just demonstrating:

"I know SQL."

You're demonstrating:

"I can take a real business problem,

work with relational data,
clean messy data,
investigate relationships,
analyze unstructured customer feedback,
use AI responsibly,
validate hypotheses statistically,
build an executive dashboard,
and recommend what the business should do."

THE PROJECT TITLE FOR YOUR PORTFOLIO

"Customer Experience & E-Commerce Performance Investigation
Using SQL, Python, Power BI & GenAI"

THE ONE-LINE RESUME VERSION

"Investigated drivers of e-commerce customer satisfaction by
combining transactional, delivery, seller, product and review
data; used SQL, Python, statistical analysis and LLM-based
sentiment/topic classification to identify operational
pain points and develop data-backed recommendations."

DATA SOURCE

Olist Brazilian E-Commerce Public Dataset

Kaggle:
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce